Unit 1

Polynomial and Rational Functions

1.1 Monomials and Polynomials

Review: Exponents

Definition

We represent repeated multiplication using **positive integer exponents**. The expression b^p is defined as

$$b^p = \underbrace{b \cdot b \cdots b}_{p \text{ copies}}$$

The number b in this expression is called the **base**, and the number p is called the **exponent**.

Note. The exponent is written as a _____ , _____ and _____ the main line of text.

Example. Consider the expression 5^3:

- The base is _____ and the exponent is _____ .
- Rewrite 5^3 as repeated multiplication: _____
- Find 5^3 by hand: _____

Note. We can add exponent _____ to any expression without changing the value, that is, for any x, $x =$ _____ .

Exponent Rules (Product and Quotient of Powers)

We have certain rules that allow us to combine expressions with exponents.

Theorem: Product of Powers Rule

For any real number x and positive integers m and n,

$$x^m x^n = x^{m+n}$$

$x^m x^n =$

Theorem: Quotient of Powers Rule

For any real number x and positive integers $m \neq n$,

$$\text{If } m > n,\ \frac{x^m}{x^n} = x^{m-n} \qquad \text{If } m < n,\ \frac{x^m}{x^n} = \frac{1}{x^{n-m}}$$

If $m > n$, $\dfrac{x^m}{x^n} =$

If $m < n$, $\dfrac{x^m}{x^n} =$

Exponent Rules (Power of a Product, Quotient, and Power)

Theorem: Power of a Power Rule

For any real number x and any positive integers n and p,

$$(x^n)^p = x^{np}$$

$(x^n)^p =$

Theorem: Power of a Product Rule

For any real numbers x and y and any positive integer n,

$$(xy)^n = x^n y^n$$

$(xy)^n =$

Theorem: Power of a Quotient Rule

For any real numbers x and $y \neq 0$ and any positive integer n,

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

$\left(\frac{x}{y}\right)^n =$

Exponent Rules (Negative Exponents and Examples)

We define the remaining integer exponents to be consistent with the product and quotient rules.

Definition: Exponent Zero

For any real number $x \neq 0$, we define

$$x^0 = 1$$

- This is consistent with the product rule:

  $x^0 x^n = x^{0+n} = \underline{\qquad} = \underline{\qquad} \cdot x^n$

- This is consistent with the quotient rule (if we allow $m = n$):

  $\dfrac{x^n}{x^n} = x^{n-n} = \underline{\qquad}$, but $\dfrac{x^n}{x^n} = \underline{\qquad}$

Definition: Negative Integer Exponents

For any real number $x \neq 0$ and positive integer n,

$$x^{-n} = \frac{1}{x^n}$$

This lets us simplify the statement of the quotient rule:

Theorem: Quotient of Powers Rule

For any real number x and integers m and n,

$$\frac{x^m}{x^n} = x^{m-n} = \frac{1}{x^{n-m}}$$

We end with some examples of applying these rules.

Example. Simplify. Assume $x \neq 0$.

$x^3 x^4 =$

$\dfrac{x^7}{x^3} =$

$\dfrac{x^5}{x^8} =$

$(x^3)^5 =$

$(x^{-3})^2 =$

$(4x)^3 =$

$\left(\dfrac{x}{3}\right)^4 =$

$\left(\dfrac{x^{-2}}{y^3}\right)^{-5} =$

Monomials

Definition: Monomial
A **monomial** is a combination of numbers and variables, put together with *multiplication only*. In simplest form, we combine all the numeric factors into a single **coefficient**, and write repeated variable factors using a single positive integer exponent. The **degree** of a monomial is the total number of variable factors, that is, the sum of all the exponents.

Example. For each expression, determine whether or not it is a monomial. If it is, write it in simplest form.

a. $3x \cdot 5xy \cdot (-6yy)$

b. $\frac{2}{3}x^3yxy^5$

c. $\dfrac{5y^3}{x}$

d. $3x^{-4}y^3$

e. 7

Example. For each monomial below, determine the coefficient and degree.

a. $4x^2$

Coefficient: _____ Degree: _____

b. $-\frac{5}{3}x^3y^5$

Coefficient: _____ Degree: _____

c. $7.4xyzw$

Coefficient: _____ Degree: _____

Polynomials

Definition: Polynomial
A **polynomial** (or **polynomial expression**) is an expression given as a sum of one or more monomials. The monomials are called the **terms** of the polynomial.

Example. For each expression below, determine whether or not it is a polynomial. If it is not, explain why not.

a. $3x^2 + 5x - 7$

b. $x - 3\sqrt{x} + 2$

c. $4x^{-2} - 3$

d. $2x^3y$

e. 5

Describing Polynomials (Part 1)

Definition

- The **degree of a polynomial** is the highest degree of any term.
- The **leading term** is the term with highest degree.
- The **leading coefficient** is the coefficient of the leading term.

If a polynomial has more than one term of the highest degree, it does not have a leading term.

Example. For each polynomial below, identify the leading term, leading coefficient, and degree.

a. $3x^3 + 2x^2 + 5x + 3$ L. term: _____ L. coeff: _____ Deg: _____

b. $x + \frac{2}{3}x^3 - 7x^5$ L. term: _____ L. coeff: _____ Deg: _____

c. $3x^2y - 4xy^2 + 1.3x^2y^2$ L. term: _____ L. coeff: _____ Deg: _____

d. $x^2 - 5xy + 6y^2$ L. term: _____ L. coeff: _____ Deg: _____

Definition

- A **polynomial in one variable** is one in which only one variable appears.
- A **homogeneous** polynomial is one in which every term has equal degree.

Example. For each polynomial below, identify it as a polynomial in one variable, a homogeneous polynomial, or neither. Also determine the degree of the polynomial.

a. $x^4 - x^2 + 2x + 1$ One variable | Homogeneous | Neither Degree: _____

b. $x^2 - 5xy - 6y^2$ One variable | Homogeneous | Neither Degree: _____

c. $6xy - 10x + 9y$ One variable | Homogeneous | Neither Degree: _____

Definition

- A polynomial is written in **descending order** if the terms are ordered from highest to lowest degree.
- A polynomial is written in **ascending order** if the terms are ordered from lowest to highest degree.

Example. Write $5x - 3x^2 - 2x^3 + 6$ in descending order.

_____

Example. Write $5y + 15 - 6y^5 + 2y^7$ in ascending order.

_____

Describing Polynomials (Part 2)

Definition

We can name polynomials by their degree. (These words are *adjectives*.)

Degree	Name	Example
0	Constant	7
1	Linear	$5x - 3$
2	Quadratic	$x^2 - 3x + 2$
3	Cubic	$x^3 - 5$

For higher degrees, we just say "of degree n."

Definition

We can name polynomials by their number of terms. (These words are *nouns*.)

Terms	Name	Examples	
1	Monomial	x^3	$2x$
2	Binomial	$x - 3$	$x^3 - 8$
3	Trinomial	$x^2 - 3x + 2$	$x^5 - x^3 + x$

For more terms, we just say "A polynomial with n terms."

Example. Describe each polynomial.

a. $x^3 - 7$ _____

b. $x^4 + x^5 - 3$ _____

c. y^2 _____

d. $2x^2 - 5xy + 2y^2$ _____

e. $x + y + z + w$ _____

f. $2x - 5$ _____

g. $x^3 - 5x^2 + 2x - 10$ _____

h. $x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4$ _____

Polynomial Functions

Definition: Polynomial Function
A function is a polynomial function if its expression is a polynomial.

Example. Let $f(x) = 2x^3 - 5x^2 + 4x + 7$. Evaluate:

a. $f(3) =$ _____

b. $f(0) =$ _____

c. $f(-2) =$ _____

Polynomials give an insight into the Order of Operations.

Definition: Order of Operations
In expressions with more than one operation, they should be completed in the following order: 1. Exponents 2. Multiplication (and division) 3. Addition (and subtraction) To violate this order, use grouping symbols (like parentheses).

The reason that this order is chosen is to make polynomials work as expected. To evaluate $f(-2)$, we:

1. found each power of -2
2. multiplied by its coefficient
3. then added up all the terms

This is why _____ come before _____ in order of operations: we want $-x^2$ to mean _____ the number x, then _____ .

Example. Evaluate each of the following.

$2^2 =$ _____ $\qquad -2^2 =$ _____ $\qquad (-2)^2 =$ _____

1.2 Polynomial Arithmetic

Adding Polynomials

Adding polynomials is an application of the associative and commutative property.

Example. Add $(2x^2 - 5x + 6) + (3x^2 - 11x - 5)$, justifying each step.

Step	Why?

We will typically not write all of these steps. If you remember that _____ signs go with their term, you may be able to directly write down the sum.

Example. Add $(3x^4 - 5x^3 + 2x + 3) + (3x^3 - 7x^2 + x - 11)$

Negating a Polynomial

It is often necessary to take the opposite of a polynomial. Recall that we can understand writing a negative sign in front of parentheses as multiplying by _____ .

Example. Simplify $-(2x^4 - 5x^3 + 6x^2 + 7x - 3)$, justifying each step.

Step	Why?

We will usually not write all of these details, either.

Example. Simplify $-(-3x^2 + 4xy - 6y^2 + 2y)$

We can think of this move as "distribute the negative sign."

Subtracting Polynomials

To subtract polynomials, we remember that subtraction is _____ the _____ .

Example. Subtract $(3x^2 - 5x + 3) - (2x^2 + 6x - 5)$, justifying each step.

Step	Why?

Typically, the only step we write out is the one where we "distribute" the negative sign.

Example. Subtract $(5x^2 - 4xy - 6y^2) - (3y^2 + 3xy - 4y^2)$

We can also think of this process as "subtracting like terms."

Example. Subtract $(5x^2 - 6x + 3) - (2x^2 + 3x - 3)$

Write:	Think:

Multiplying a Monomial and a Polynomial

Multiplying a monomial and a polynomial is an application of the _____ property, together with exponent rules.

Example. Multiply $3x^2y(4x^2 - 6xy + 3y^2)$, justifying each step.

Step	Why?

We will typically write fewer steps here, as well.

Example. Multiply $-2x(-x^2 + 7x - 3)$

Multiplying Polynomials: Introduction

To multiply two polynomials (with more than one term), we still apply the _____ property, but we apply it _____ .

Example. Multiply $(x^2 - 3x + 2)(2x - 1)$, justifying each step.

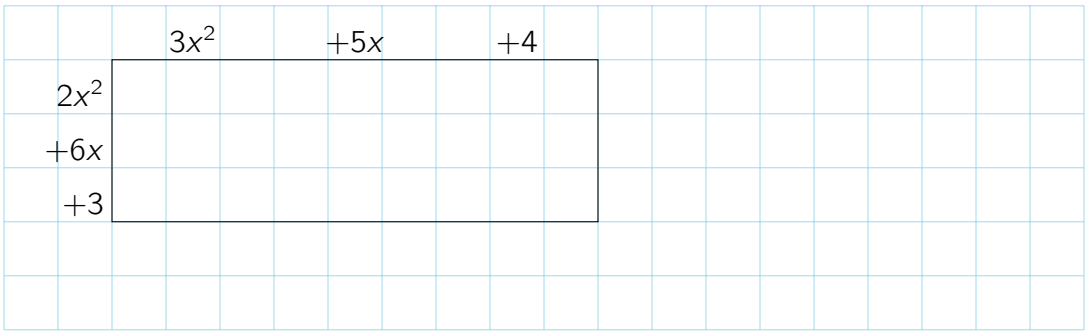
Step	Why?

The result is to multiply every _____ in the first polynomial by every _____ in the second. In the next two videos, we'll see two ways to arrange this.

Multiplying Polynomials: Rectangle Method

We can think of multiplication as describing the _____ of a rectangle.

Example. Multiply $(3x^2 + 5x + 4)(2x^2 + 6x + 3)$ by interpreting the polynomials as describing the sides of a rectangle.



Example. Multiply $(4x + 3y - 6)(3x - 5y + 2)$.

Multiplying Polynomials: FOIL Method

When we multiply two _____ , there are _____ possible products.

Example. Multiply $(A + B)(C + D)$ as a rectangle.

The _____ possible products are:

F Multiply the **F**irst terms of each binomial
O Multiply the **O**uter pair of the product
I Multiply the **I**nner pair of the product
L Multiply the **L**ast terms of each binomial

Using this as a shortcut to multiply binomials is called the **FOIL method**. Remember, we can only use this to multiply two _____ .

Example. Multiply $(2x - 3)(x + 5)$ as a rectangle and using the FOIL method.

When we multiply two _____ binomials, we know we will combine the middle terms. With practice, we may skip writing this step.

Example. Multiply $(2x - 3)(3x - 2)$ using the FOIL method.

Multiplying Three Polynomials

If we wish to multiply three or more polynomials, we do it the same we do with numbers: multiply the _____ , then multiply _____ .

Example. Multiply $(x-4)(x+2)(x-1)$.

First we multiply _____ using the _____ .

Now we must multiply _____ .

$(x-4)(x+2)(x-1) =$ _____

Powers of Polynomials

We take powers of polynomials by _____ .

Example. Find $(2x - 3y)^2$

Example. Find $(x + 3)^3$

Example. Find $(x^2 - 2x + 2)^2$

1.3 Factoring: Basic Methods

Factoring Numbers Review

Let's review the notion of factoring integers.

Definition

- A **factor** (noun) of an integer n is an integer k such that $n = kd$ for some integer d.
- To **factor** (verb) an integer is to determine its factors.
- A **prime number** p is an integer whose only factors are $\pm p$ or ± 1.
- The **prime factorization** of an integer as a product of prime numbers.
- We can find the prime factorization of an integer by making a **factor tree**.

Example. Make a factor tree to find the prime factorization of the number 18.

Theorem: Fundamental Theorem of Arithmetic

Any integer can be written uniquely as a product of prime numbers.

Note. For purposes of "uniqueness," we ignore negative signs and order; so $10 = 2 \cdot 5 = 5 \cdot 2 = (-2) \cdot (-5)$ would be considered "the same" factorization.

Introduction to Factoring Polynomials

When we consider factoring polynomials, we will consider polynomials with *integer* coefficients.

Definition
• A **factor** of an integer n is a number k such that $n = kd$ for some integer d. • A **factor** of a monomial M is a monomial A such that $M = AB$ for some monomial B. • A **factor** of a polynomial P is a polynomial Q such that $P = QS$ for some polynomial S.

Note that these definitions are very similar!

Example.

As an integer, 5 is a factor of 30 because $30 = 5 \cdot$ _____.

As a monomial, $3x^2$ is a factor of $9x^3y$ because $9x^3y = (3x^2)($_____$)$.

As a polynomial, $4x+3$ is a factor of $4x^2+3x$ because $4x^2+3x = ($_____$)(4x+3)$.

When we factor (verb) a polynomial, our goal is to rewrite it in a factored form.

Definition
An expression is in "factored form" if it is written as a product of simpler expressions, that is, if the last step to evaluate it would be _____ .

Example. For each expression below, determine whether it is in a factored form.

a. $3x^2y$ _____

b. $(2x-3)(5x+2)$ _____

c. $3x(x+5)-7(x+5)$ _____

d. $4x^2+6x-3$ _____

e. $x^2(3x^2-4x-4)$ _____

f. $3x(5-3x)+4$ _____

Note that _____ are automatically in a factored form.

Greatest Common Factor

Definition: Greatest Common Factor (integers)
A **common factor** of two (or more) integers is a number that is a factor of both (or all). The **greatest common factor** (or **GCF**) of two (or more) integers is their largest (positive) common factor.

Example. List all (positive) factors of the integers 8 and 20. Then, give their GCF.

- Factors of 8 _____
- Factors of 20 _____

The largest number on both lists is _____ , so this is the greatest common factor of 8 and 20.

We can also use a calculator or computer to find the GCF of two numbers.

Definition: Greatest Common Factor (monomials)
A **common factor** of two (or more) monomials is a monomial that is a factor of both (or all). The **greatest common factor** (or **GCF**) of two (or more) monomial is their common factor that has: • the largest possible coefficient • the highest possible degree

To find the GCF of monomials, we find:

- the GCF of the coefficients
- the lowest power of each variable that appears in the monomials

If we take the prime factorization of each coefficient, we can treat prime numbers like variables. Or, we can use a calculator to find the GCF of the coefficients.

Example. Find the GCF of $3x^4y^3$, $6x^3y^5$, and $9x^2y^7$.

- For the coefficient: _____
- For x: _____
- For y: _____

The GCF of $3x^4y^3$, $6x^3y^5$, and $9x^2y^7$ is _____ .

Factoring Out the GCF

The first step in factoring any polynomial is to find any monomial factor. We call this **factoring out the GCF**. Let's see an example, then summarize the process.

Example. Factor out the GCF from the polynomial $6x^5y^2 - 18x^4y^3 + 12y^4$

Procedure: Factoring Out the GCF

1. Find the GCF of the polynomial's terms.
2. Divide each term by the GCF to determine the remaining factor, using properties of exponents.
3. Write the original polynomial as the GCF multiplied by the polynomial made up of the remaining factors.
4. To check, note that:
   (a) The remaining factors have no common factors.
   (b) When you multiply, you get the original polynomial.

A Negative GCF

It is more convenient to work with polynomials that have a _____ leading term. If the leading term is _____ , we can fix this by factoring out a negative GCF.

Example. Factor out the GCF from $-5x^5 + 20x^4 - 25x^3$ so that the leading term of the polynomial factor is positive.

Sometimes, we simply factor out -1.

Example. Factor out the GCF from $-3x^2 - 7x + 9$ so that the leading term of the polynomial factor is positive.

Factoring by Grouping

Sometimes, when we multiply two binomials, we get a polynomial in four terms. To factor such a polynomial, we want to get the two binomials back.

Example. Multiply $(2x-3)(3y+1)$ by applying the distributive property twice.

$(2x-3)(3y+4) =$

Note that in this example, there are no _____ to combine.

If we want to get back to the binomial factors, we can apply this process backward!

Example. Factor $6xy - 9y + 8x - 12$ as the product of two binomials.

Step	Why?

Procedure: Factor by Grouping

To factor a polynomial with four terms by grouping:

1. Use the associative property of addition to "group" the terms into two pairs.
2. For each pair of terms, factor out the GCF.
3. Note that the remaining binomial factors for each pair are the same.
4. Factor out the remaining binomial factors to write the polynomial as a product of two binomials.
5. To check, multiply the two binomials, and confirm you get the original polynomial back.

Note that it is important that, when you create the groups, the operation combining them is *addition*.

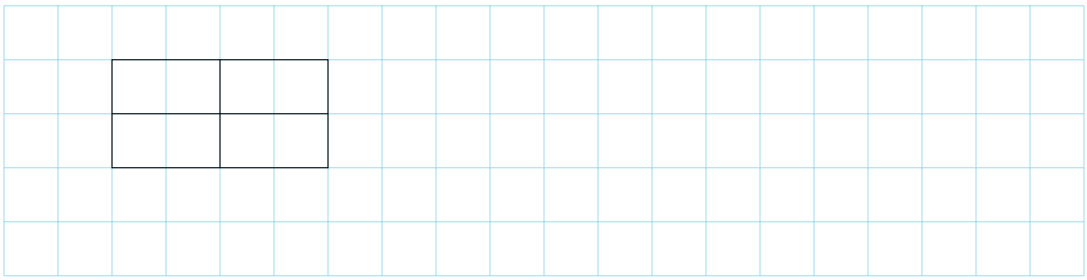
Example. Factor $2y^3 - 3y^2 - 10y + 15$ by grouping.

$2y^2 - 3y^2 - 10y + 15 =$

Factoring by Grouping Rectangle Method

We can also perform factoring by grouping by running the rectangle method of multiplication backward. This can make it easier to keep track of signs.

Example. Factor $p^2 + 3ps - 2pq - 6qs$ by grouping, using the rectangle method.



Example. Factor $2x^5 - x^3 + 10x^2 - 5$ by grouping, using the rectangle method.

1.4 Factoring: Trinomials and Special Forms

Factoring Quadratic Trinomials: the AC Method

When we multiply two linear binomials (in one variable or homogeneous in the same two variables), the result is a quadratic trinomial.

Example. Multiply $(2x-3)(x+2)$, using the rectangle method and the FOIL shortcut.

When we multiply, we must _____ . When we factor, we must reverse this, i.e. "_____" the middle terms.

Observing the multiplication, we see that the middle terms (to be combined) have the same _____ as the first and last terms. This gives us the following technique.

Procedure: The AC Method

To factor a quadratic trinomial:

1. Multiply the coefficients of the first and last terms.
2. Find two numbers that have the same product, and whose sum is the coefficient of the middle term.
3. Rewrite the middle term as a sum of two terms using these coefficients.
4. The polynomial now has four terms. Factor by grouping.

If we write a quadratic trinomial as $ax^2 + bx + c$ (or $ax^2 + bxy + cy^2$), the two numbers we seek in step 2 have product ac and sum b.

Example. Factor $2x^2 + x - 6$ using the AC Method.

Example. Factor $6x^2 - 11xy + 3y^2$ using the AC Method.

AC Method with the Calculator

When using the AC Method, we often need to find all factors of a large number with many factors.

Example. Use the AC Method to factor $6x^2 + 23xy + 20y^2$ using the AC Method.

We must find two numbers whose product is _____ and whose sum is _____ . This requires we find all factors of _____ !

1	×	120	= 120		1	+	120	=	121
2	×	60	= 120		2	+	60	=	62
3	×	40	= 120		3	+	40	=	43
4	×	30	= 120		4	+	30	=	34
5	×	24	= 120		5	+	24	=	29
6	×	20	= 120		6	+	20	=	26
8	×	15	= 120		8	+	15	=	23
10	×	12	= 120		10	+	12	=	22

This is a lot of factors! Whenever we want to do highly repetative arithmetic, we want a _____ to do it for us. The procedure here uses the TI-84 calculator, but it can be done with many other calculators or with a computer.

Procedure: Factors for the AC Method on the TI-84

1. Press [Y=] to access the function entry screen.
2. Enter `##/X` as Y_1 and `##/X+X` as Y_2, where `##` is the desired product.
3. Press [2ND] [GRAPH] to access the "Table" feature.
4. The `X` and Y_1 columns show the two factors, and the Y_2 column shows their sum. Use the up and down arrow keys to scroll through the table until the desired sum is found.

Example. Factor $12x^2 - 29x - 60$ using the AC Method, using a calculator to help split the middle term.

Quadratic Trinomials with Leading Coefficient 1

In general, when we use the AC Method, the numbers used to split the middle term do not appear directly in the factors. However, look what happens when the leading coefficient is 1.

Example. Factor using the AC Method.

$x^2 - 5x + 6 =$

$x^2 + 4xy - 12y^2 =$

We notice that the numbers that appear in the factors are _____ as the ones we found in the AC Method. This gives us a shortcut, as long as the leading coefficient (in at least one variable) is 1.

Example. Determine whether the shortcut for the AC Method can be used. Factor using the shortcut if possible.

$2x^2 + 5x - 3$

$x^2 + 3x - 18$

Prime Quadratic Polynomials

Recall the definition of a prime number.

Definition: Prime Number
An integer p is **prime** if its only factors are ± 1 and $\pm p$.

The concept is that a prime number "cannot be factored." We have a parallel definition for polynomials.

Definition: Prime Polynomial
A polynomial with integer coefficients is said to be **prime** if it cannot be written as a product of polynomials with integer coefficients of lower degree or with smaller coefficients.

We can always factor out ± 1 from any polynomial; it is prime if we can't factor out anything else. How can we recognize a prime polynomial?

Assuming no greatest common factor (except ± 1), the following polynomials are prime.

- Any linear polynomial.
- Any quadratic polynomial with four terms for which grouping fails.
- Any quadratic trinomial for which the AC Method fails.

Example. Identify each polynomial as prime or not prime. If it is not prime, factor.

$x^2 - 4xy + 3x - 12y$

$x^2 - 4xy + 5y^2$

$2x^2 - 7x + 6$

$xy - 5x - 3y + 7$

Square and Cube Numbers

In factoring, it can be useful to recognize certain numbers as "perfect squares" and "perfect cubes," that is, powers of integers.

x	1	2	3	4	5	6	7	8	9
x^2	1	4	9	16	25	36	49	64	81
x^3	1	8	27	64	125	216	343	512	729

If you spend a lot of time factoring, you may memorize these lists. If you don't have a list like this memorized, you can use a calculator or computer to make one.

Procedure: List Squares and Cubes on the TI-84

1. Press [Y=] to access the function entry screen.
2. Enter $\mathtt{X}^2$ as $\mathtt{Y}_1$ and $\mathtt{X}^3$ as $\mathtt{Y}_2$.
3. Press [2ND] [GRAPH] to access the "Table" feature.
4. The $\mathtt{Y}_1$ column shows perfect squares, and the $\mathtt{Y}_2$ column shows perfect cubes. Use the up and down arrow keys to scroll through the table to see if a particular number appears.

Example. Use the table above or a table generated on the calculator to identify each of the following numbers as a perfect square, a perfect cube, both, or neither.

1. 20 Perfect square Perfect cube Neither
2. 49 Perfect square Perfect cube Neither
3. 125 Perfect square Perfect cube Neither
4. 64 Perfect square Perfect cube Neither
5. 400 Perfect square Perfect cube Neither
6. 200 Perfect square Perfect cube Neither

Special Forms: Perfect Squares

To help us to factor, there are certain special products that it is worthwhile to memorize. The first form we memorize is the perfect square of a binomial.

To see these forms, find each product using the FOIL shortcut.

$(A+B)^2 =$

$(A-B)^2 =$

Recognizing a Perfect Square

A trinomial is a **perfect square trinomial** when:

- Two terms are positive perfect square monomials (_____ a square number, and _____ powers of variables) $+A^2$ and $+B^2$.
- The third term is $+2AB$ or $-2AB$.

Example. For each polynomial below:

a. Determine whether two terms are perfect squares.
b. Determine whether the trinomial is a perfect square.
c. If it is, factor it as such.

$4x^4 - 12x^2 + 9$

a.

b.

c.

$x^2 + 8xy + 16y^2$

a.

b.

c.

$x^2 - 5xy + 4y^2$

a.

b.

c.

$x^2 - 6x - 9$

a.

b.

c.

Special Forms: Difference of Squares

Example. Factor $x^2 - 9$ using the AC Method.

- To apply the AC Method, we need to add the term _____ .

We can generalize this to a form called **difference of squares** by multiplying:

$(A - B)(A + B) =$

Recognizing a Difference of Squares
A binomial is a **difference of squares** when: • Each term is a perfect square monomial (ignoring signs). • One is positive, and the other negative.

Example. For each polynomial below:

a. Determine whether it is a difference of squares.
b. If it is, factor it as such.

$9x^2 - 16$

a.

b.

$x^4y^2 - 36z^6$

a.

b.

$z^4 + 81$

a.

b.

$x^3 - y^2$

a.

b.

A Sum of Squares is Prime

What happens in the difference of squares form if we replace B by $-C$?

- $A^2 - B^2 = A^2 - (_____)^2 = _____$
- $(A-B)(A+B) = (A-(_____))(A+(_____)) = _____$

This is the _____ form!

Example. Try to factor $x^2 + 9$ using the AC Method. Show that it is prime.

Note that $x^2 + 9 = x^2 + \qquad + 9$.

1	×	9	=	9		1	+	9	=										
3	×	3	=	9		3	+	3	=										
−1	×	−9	=	9		−1	+	−9	=										
−3	×	−3	=	9		−3	+	−3	=										

Note that any two numbers whose product is $+9$ must have the same _____ , so their sum can never be _____ .

Theorem
As long as we are limited to polynomails with real number coefficients, any polynomial of the form $A^2 + B^2$ cannot be factored as a product of two binomials.

Special Forms: Difference of Cubes

Let's see what happens when we multiply $(A-B)(A^2+AB+B^2)$ using the rectangle method.

Notice that all the _____ cancel out! This gives us a special form called a **difference of cubes**.

Recognizing a Difference of Cubes
A binomial is a **difference of cubes** when: • Each term is a perfect cube monomials (_____ a cubed number, and powers of variables multiples of _____). • One is positive, the other negative.

Example. Factor each difference of cubes.

$x^3 - 27 =$

$8x^6y^3 - 27 =$

Note. We can generalize this for *any* powers (not just 2 and 3) but we will only consider 2 and 3 for this class.

Special Forms: Sum of Cubes

What happens in the difference of cubes form if we replace B by $-C$?

- $A^3 - B^3 = A^3 - (_____)^3 =$ _____
- $(A - B)(A^2 + AB + B^2) = (A - (_____))(A^2 + A(_____) + (_____)^2)$

  $=$ _____

This is a _____ form, the _____ **of cubes**.

Recognizing a Sum of Cubes
A binomial is a **sum of cubes** when: • Each term is a perfect cube monomials (_____ a cubed number, and powers of variables multiples of _____). • Both terms are positive.

Note. If both terms are _____ , we can factor out _____ to get a sum of cubes.

Example. Factor each sum of cubes.

$8x^3 + 1 =$

$x^6 + y^6 =$

Note. We can generalize this for any _____ powers (not just 3). For _____ powers, the sum is prime.

Using Special Forms

We now have five special forms of polynomials, that we can use to help us factor.

Perfect Square Trinomials
$(A+B)^2 = A^2 + 2AB + B^2 \qquad (A-B)^2 = A^2 - 2AB + B^2$

Difference of Squares
$A^2 - B^2 = (A-B)(A+B)$

Sum and Difference of Cubes
$A^3 - B^3 = (A-B)(A^2 + AB + B^2) \qquad A^3 + B^3 = (A+B)(A^2 - AB + B^2)$

How do we use this? We should be in the habit, when factoring, of being on the lookout for these special forms.

- When factoring a trinomial, two _____ terms should "jump out" at us, and we should check for a perfect square.
- When factoring a _____ , we can *only* factor if we have one of the special forms!

Example. For each binomial, determine whether it is a special form, and if so, factor it.

$x^4 - 1$

$8y^6 + 343$

$5x^4 - 1$

$25y^2 + 16$

$x^3 - 25$

$x^6 - 64$

Note. If a binomial is *both* a difference of squares and of cubes, we'll factor it first as a difference of _____ .

1.5 Factoring Fully and Polynomial Equations

Factoring Fully

Definition: Factoring Fully

A polynomial is **fully factored** if:

- The greatest common factor, if any, has been factored out.
- All polynomial factors are prime.

Example. Fully factor the polynomial $x^4y^2 - 16y^2$

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Factoring Trinomials

We know how to factor quadratic trinomials. When they have higher degree, we may be able to pretend they're quadratic.

Example. Fully factor $10x^4 - 25x^2 - 60$.

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Disguised Difference of Squares

Sometimes, we have a difference of squares in which one or both squares is a _____ .

Example. Fully factor the polynomial $x^4 - 10x^3 + 25x^2 - 36$.

1. Factor out the GCF
2. Factor the polynomial factor
3. Factor each new polynomial factor
4. Put it all together

We can check for this whenever we have 3 terms that are _____ .

Example. Fully factor the polynomial $x^4 - 4x^2 + 12x - 9$.

1. Factor out the GCF
2. Factor the polynomial factor
3. Factor each new polynomial factor
4. Put it all together

Multiple Iterations

We must continue factoring until all remaining factors are prime.

Example. Fully factor $5x^6 + 15x^5 - 5x^2 - 15x$.

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Factor Any Polynomial

Follow these steps to factor any polynomial.

1. Factor out the GCF.

2. The next step depends on the number of terms:

   Two Terms Can only factor if it's a special form.

   (a) For degree 2 or 3, anything other than $A^2 - B^2$, $A^3 - B^3$, or $A^3 + B^3$ is prime.

   (b) For degree 4+, may need extra special forms; if this fails, turn to advanced techniques.

   Three Terms Try these things.

   (a) If it's quadratic, use the AC Method. If this fails, it's prime.

   (b) If it's higher degree, try to pretend it's quadratic and use the AC Method. If this fails, turn to advanced techniques.

   Four Terms Try these things.

   (a) Factor by grouping. If this fails and it's quadratic, can't factor as product of binomials.

   (b) If grouping fails, try the difference of squares trick.

   (c) If both fail, turn to advanced techniques.

   Five+ Terms Turn to advanced techniques.

3. After factoring, return to step 2 with each factor.

1.6 Solving Polynomial Equations

Polynomial Equations

Definition: Polynomial Equation
An equation is a **polynomial equation** if the expressions on either side are polynomials, at least one of which is degree 2 or higher.

Example. $x^2 - 5x - 14 = 0$ is a polynomial equation. Determine whether each value below is a solution to this equation:

a. $x = 7$ _____

b. $x = 2$ _____

c. $x = -2$ _____

Example. $(x-5)(x+3) = 9$ is a polynomial equation. Determine whether each value below is a solution to the equation:

a. $x = 6$ _____

b. $x = 3$ _____

c. $x = -4$ _____

Note that polynomial equations typically have _____ solution.

Polynomial Equations in Factored Form

Zero Property of Multiplication
For any real numbers a and b, if $ab = 0$ then $a = 0$ or $b = 0$

We can use this fact to solve polynomial equations.

Example. Use the Zero Property of Multiplication to solve the polynomial equation

$$(2x - 1)(x + 3)(x - 5) = 0$$

Since the _____ is _____ , one of the _____ is _____ .

We can check that these really are solutions to the original equation:

Note. We will solve *many* equations of the form $ax + b = 0$ as we continue to work with polynomials! The solution is always $x = -\frac{b}{a}$.

Polynomial Equations in Standard Form

Definition: Standard Form
A polynomial equation is said to be in **standard form** if it has an unfactored polynomial on one side, and zero on the other.

Given a polynomial equation in standard form, we first factor the polynomial.

Example. Solve $6x^4 + 14x^3 - 40x^2 = 0$.

Factor $6x^4 + 14x^3 - 40x^2$

Solve:

Check each solution

Polynomial Equations in General Forms

Procedure: Solving a Polynomial Equation
1. Write the equation in standard form. This may involve doing multiplication and/or adding or subtracting on both sides. 2. Once the polynomial is in standard form, factor the polynomial side. 3. Use the Zero Property of Multiplication to solve.

Note. If the polynomial is given in the form factored form $= 0$, there is no need to multiply out, then factor again. However, if it is in the form factored form $= c$ for some $c \neq 0$, it *is* necessary to multiply out, subtract c from both sides, then factor again. The eventual factors will be different.

Example. Solve $(x-3)(x-5) = 8$

1. Write in Standard Form

2. Factor:

3. Set each factor equal to zero

Check each solution

Example. Solve $x^3 - 4x^2 + 8 = x^3 - 5x^2 + 3x + 6$

1.

2.

3.

Check:

Solve Any Equation

Solving polynomial equations uses a whole new technique! (New steps in red.)

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - If the function is not one-to-one, there may be many solutions.
  - If the function's range is not all real numbers, there may be no solutions.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

1.7 Function Algebra

Function Algebra

We have learned one way to combine functions: _____ . We can also combine them the way we do numbers.

Definition: Sum, Difference, Product, Quotient of Functions

For any functions f and g:

- Their **sum** $f+g$ is given by $(f+g)(x) = f(x) + g(x)$
- Their **difference** $f-g$ is given by $(f-g)(x) = f(x) - g(x)$
- Their **product** $f \cdot g$ is given by $(f \cdot g)(x) = f(x) \cdot g(x)$
- Their **quotient** $\frac{f}{g}$ is given by $\left(\frac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$

Note. We may write fg rather than $f \cdot g$. It is important to distinguish between $f \cdot g$ and $f \circ g$ (the composition).

Example. Let $f(x) = 3x^2 - 5x + 2$, $g(x) = 2\,|x-5|$. Find each of the following:

$f(3) =$

$g(3) =$

$(f+g)(3) =$

$(f-g)(3) =$

$(f \cdot g)(3) =$

$\left(\frac{f}{g}\right)(3) =$

Constructing new functions in this way is referred to as "function algebra."

Domains of New Functions

For x to be in the domain of the sum, difference, product, or quotient of two functions f and g, we need to be able to do three things:

- Evaluate $f(x)$
- Evaluate $g(x)$
- Perform the appropriate operation (addition, subtraction, multiplication, or division).

We can add, subtract, or multiply _____ two numbers, but we can't divide by _____ .

Domain of Sum, Difference, Product, and Quotient
For any two functions f and g, $$\begin{aligned} \text{Dom}(f+g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \\ \text{Dom}(f-g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \\ \text{Dom}(f \cdot g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \end{aligned}$$ $$\text{Dom}\left(\frac{f}{g}\right) = (\text{Dom}\, f) \cap (\text{Dom}\, g) \cap \{x \mid g(x) \neq 0\}$$

Example. Suppose $\text{Dom}\, f = (-\infty, 3]$ and $\text{Dom}\, g = (-6, 15)$. Find each domain:

$\text{Dom}\,(f+g) =$

$\text{Dom}\,(f-g) =$

$\text{Dom}\,(f \cdot g) =$

Suppose also the solution set of $g(x) = 0$ is $\{-2, 1\}$. Find the domain:

$\text{Dom}\left(\frac{f}{g}\right) =$

Examples from Pairs

Example. Define the following functions:

$$\begin{aligned} f &= \{(-8,\ \ 8),\ (-6,-3),\ (3,\ \ 0),\ (4,\ \ 3),\ (9,-3)\} \\ g &= \{(-8,-6),\ (-4,\ \ 5),\ (3,-7),\ (7,-3),\ (9,\ \ 0)\} \\ h &= \{(-8,\ \ 3),\ (\ \ 0,-9),\ (1,-8),\ (3,-3),\ (9,\ \ 7)\} \end{aligned}$$

Find each of the following.

a. $(f+g)(3)$ _____

b. $(h-g)(1)$ _____

c. $(f \cdot g)(9)$ _____

d. $\left(\frac{g}{h}\right)(-8)$ _____

e. $\left(\frac{f}{h}\right)(3)$ _____

Example. Define the following functions:

$$\begin{aligned} f &= \{(-8,\ \ 8),\ (-6,-3),\ (3,\ \ 0),\ (4,\ \ 3),\ (9,-3)\} \\ g &= \{(-8,-6),\ (-4,\ \ 5),\ (3,-7),\ (7,-3),\ (9,\ \ 0)\} \\ h &= \{(-8,\ \ 3),\ (\ \ 0,-9),\ (1,-8),\ (3,-3),\ (9,\ \ 7)\} \end{aligned}$$

Find each of the following.

a. $\operatorname{Dom} f$ _____

b. $\operatorname{Dom} g$ _____

c. $\operatorname{Dom} h$ _____

d. $\operatorname{Dom}(f+g)$ _____

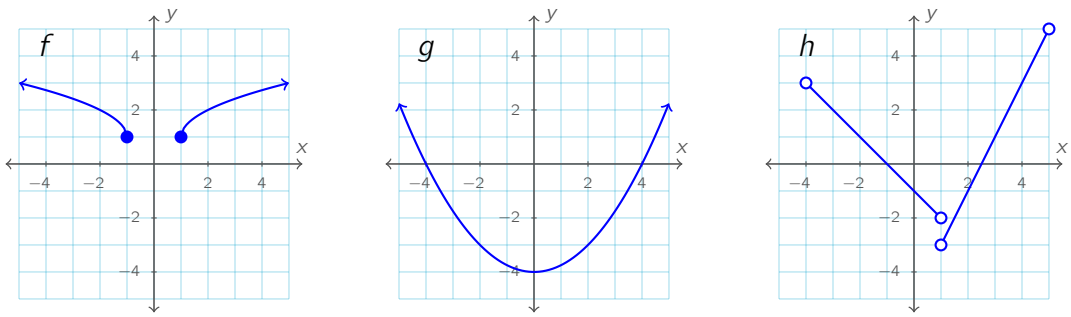
e. $\operatorname{Dom}(f-h)$ _____

f. $\operatorname{Dom}(g \cdot h)$ _____

g. $\operatorname{Dom}\left(\frac{h}{g}\right)$ _____

Examples from Graphs

Example. The functions f, g, and h are graphed below.



Find each of the following:

a. $(f+g)(2)$ _____

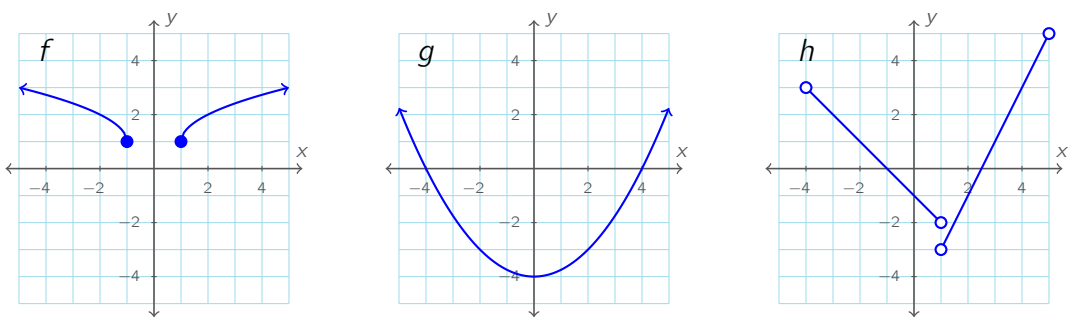
b. $(g-h)(4)$ _____

c. $(f \cdot h)(1)$ _____

d. $\left(\frac{f}{h}\right)(-2)$ _____

e. $\left(\frac{h}{g}\right)(4)$ _____

Example. The functions f, g, and h are graphed below.



Find each of the following:

a. Dom f _____

b. Dom g _____

c. Dom h _____

d. Dom$(f+h)$ _____

e. Dom $\left(\frac{g}{f}\right)$ _____

Examples from Equations

Example. Define the following functions.

$f(x) = 2x - 3$ $\qquad$ $g(x) = x^2 - 3x - 10$ $\qquad$ $h(x) = x^2 - 9$

Find each of the following.

a. $(f + g)(2)$ _____

b. $(f - h)(-1)$ _____

c. $(g \cdot h)(0)$ _____

d. $\left(\frac{h}{g}\right)(5)$ _____

e. $\left(\frac{f}{g}\right)(-4)$ _____

Example. Define the following functions.

$f(x) = 2x - 3$ $\qquad$ $g(x) = x^2 - 3x - 10$ $\qquad$ $h(x) = x^2 - 9$

Find expressions for the following. Simplify as much as possible.

a. $(f + g)(x) =$

b. $(h - g)(x) =$

c. $(g \cdot h)(x) =$

Write an expression for the following. Do not simplify.

d. $\left(\frac{g}{h}\right)(x) =$

Polynomials and Function Algebra

Theorem
Polynomials are exactly those functions that can be built from linear and constant functions by addition, subtraction, and multiplication.

Example. Show how $P(x) = 2x^2 + 2x - 7$ can be built from linear and constant functions by addition, subtraction, and multiplication.

Note. There are many other ways we could break this function apart.

Theorem
The domain of any polynomial function is _____ . In notation, if P is a polynomial function, $$\text{Dom } P = ____ = _____$$

Studying the arithmetic of polynomial expressions, we discovered the following fact about polynomial functions.

Theorem: Closure of Polynomials
The set of polynomial functions is **closed** under addition, subtraction, and multiplication. That is, if P and Q are polynomial functions, $P + Q$, $P - Q$, and $P \cdot Q$ are also polynomial functions.

The same is not true for _____ .

1.8 Rational Functions

Rational Expressions

Definition: Rational Expression
A **rational expression** is a fraction whose numerator and denominator are both polynomials.

Note. Any polynomial is a rational expression with denominator _____ . This is like the relationship between _____ and _____ .

Example. Determine whether each of the following are rational expressions.

a. $\dfrac{3x^2 - 5x + 7}{2x^2 + 4x - 11}$

b. $\dfrac{1}{x^2 - 5x - 2}$

c. $4x^5 - 3$

d. $\dfrac{3x^2}{2y}$

e. $\dfrac{2x + 5}{\sqrt{3x - 7}}$

f. $\dfrac{|x^2 + 5x + 4|}{2x - 7}$

g. $\dfrac{x^2 - x^{1.5} + 3x}{x^{2.5} - 3}$

Review: Simplifying Fractions

Before we manipulate rational expressions, we must remember how we manipulate fractions. Recall the following definitions:

Definition
• Two fractions are **equivalent** if they represent the same number. • A fraction is in **lowest terms** if its numerator and denominator are as small as possible and it includes at most one negative sign. • To **simplify** a fraction is to write an equivalent in lowest terms.

We can simplify a fraction by "canceling" any common factors of the numerator and denominator.

Example. Simplify the following fractions.

a. $\dfrac{12}{16} =$

b. $\dfrac{-15}{-10} =$

c. $\dfrac{18}{-21} =$

When we deal with larger numbers, this process is easier if we factor our numerator and denominator into primes.

Example. Simplify the fraction $\dfrac{252}{540}$ by factoring the numerator and denominator into primes.

Simplifying Ratios of Monomials

To simplify a rational expression in which the numerator and denominator are both monomials, we use the properties of exponents, specifically:

Theorem: Quotient of Powers Rule
For any real number x and positive integers $m \neq n$, $$\text{If } m > n,\ \frac{x^m}{x^n} = x^{m-n} \qquad \text{If } m < n,\ \frac{x^m}{x^n} = \frac{1}{x^{n-m}}$$

We can apply this to each variable.

Example. Simplify.

a. $\dfrac{x^2y^3z}{x^3yz^5} =$

b. $\dfrac{3a^2b^5c}{9a^2c^4d^2} =$

While they aren't actually monomials, we can also do this with expressions that include negative exponents.

Example. Simplify. Give your answer without using negative exponents.

$\dfrac{x^{-3}y^2z^5}{x^2y^{-4}z^3} =$

Note that if a variable appears only once with a negative exponent, we can write it with a positive exponent by "moving" it between the numerator and denominator.

Example. Simplify. Give your answer without using negative exponents.

$\dfrac{4a^{-3}b}{5b^{-2}c^{-1}} =$

Simplifying Ratios of Polynomials

Simplifying with monomials was easy, because they are automatically factored. With polynomials, just like with numbers, we must factor before we can simplify a rational expression.

Example. Simplify $\dfrac{2x^3 - 7x^2 + 6x}{2x^3 - 16}$.

To do this, we first need to fully factor the numerator and denominator.

$2x^3 - 7x^2 + 6x =$

$2x^3 - 16 =$

Then, we rewrite with the numerator and denominator in factored form, and cancel any common factors.

$$\frac{2x^3 - 7x^2 + 6x}{2x^3 - 16} =$$

To finish, we may or may not multiply any remaining factors. Often, we will leave our answer in factored form.

Example. Simplify $\dfrac{2x^3 + 3x^2 - 18x - 27}{4x^3 - 12x^2 - 9x + 27}$.

Term Order in Factors

Sometimes, factors appear to be common, but the order of terms will be different.

Example. Simplify $\dfrac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32}$.

To do this, we first need to fully factor the numerator and denominator.

$16 + 12x - 4x^2 - 3x^3 =$

$9x^3 - 18x^2 - 16x + 32 =$

Then, we rewrite with the numerator and denominator in factored form.

$$\frac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32} =$$

We see two pairs of factors that are not *quite* the same.

- _____ and _____

- _____ and _____

Using this to rewrite:

$$\frac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32} =$$

Rational Functions

Definition: Rational Function

We can define a rational function in two ways.

- A rational function is one whose expression is a rational expression.
- A rational function is one that can be found by dividing two polynomial functions, that is, if P and Q are polynomial functions, $R = \dfrac{P}{Q}$ is a rational function.

Example. Let $R(x) = \dfrac{2x^2 - 7x + 5}{3x^2 - 7x + 10}$. Evaluate each of the following, if defined.

a. $R(4) =$

b. $R(-2) =$

c. $R(0) =$

As we study the arithmetic of rational expressions, we will discover the following fact.

Theorem: Closure of Rational Functions

The set of rational functions is **closed** under addition, subtraction, multiplication, and division. That is, if R and S are rational functions, $R + S$, $R - S$, $R \cdot S$, and $\dfrac{R}{S}$ are also rational functions.

Domain of a Rational Function

Recall that we get a rational function by dividing two polynomial functions, and that the domain of a polynomial function is all real numbers. This tells us how to find the domain of any rational function.

Theorem: Domain of a Rational Function
The domain of a rational function $R(x) = \dfrac{P(x)}{Q(x)}$ is $\{x \mid Q(x) \neq 0\}$.

We can also see this directly: we can plug any number we like into a polynomial, but we can't _____ by _____ .

Example. Find the domain of the rational function $f(x) = \dfrac{3x-5}{2x+1}$.

To do this we must solve

Dom f =

Example. Find the domain of the rational function $g(x) = \dfrac{2x-3}{x^2-7x+12}$.

To do this we must solve

Dom g =

Simplification and Domain Restrictions

When we simplify the expression for a rational function, *the domain stays the same*. Sometimes, we must note a domain restriction.

Example. Let $h(x) = \dfrac{3x^2 - 2x - 8}{x^2 + 4x - 12}$.

a. Find the domain of h.

b. Simplify the expression for h.

c. Write the simplified form of h, noting domain restrictions.

1.9 Arithmetic of Rational Expressions

Review: Multiplying and Dividing Fractions

Multiplying and dividing fractions is straightforward.

Procedure: Multiplying Fractions
To multiply fractions, multiply the numerators, then multiply the denominators.

Example. Multiply $\frac{3}{8} \cdot \frac{10}{9}$. Simplify your answer as much as possible.

$$\frac{3}{8} \cdot \frac{10}{9} =$$

Note. When we simplify, we must _____ the numerator and denominator. We didn't need to multiply _____ or _____ ! In fact, we could have just factored _____ .

Procedure: Dividing Fractions
To divide fractions, rewrite as multiplication by the reciprocal. That is, for any fractions $\frac{a}{b}$ and $\frac{c}{d} \neq 0$, $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$.

Example. Divide $\frac{7}{9} \div \frac{14}{15}$. Simplify your answer as much as possible.

$$\frac{7}{9} \div \frac{14}{15} =$$

If we want to include integers, we write them with denominator _____ .

Example. Find each of the following.

a. $\frac{5}{6} \cdot 9 =$

b. $\frac{4}{5} \div 8 =$

Multiplying Rational Expressions

We multiply rational expressions the same way we do fractions.

Example. Multiply. Simplify your answer as much as possible.

$$\frac{5x^2 - 25x}{x^2 - 3x + 2} \cdot \frac{x^3 - 8}{x^2 - 25} =$$

$$\frac{5x^2 - 25x}{x^2 - 3x + 2} \cdot \frac{x^3 - 8}{x^2 - 25} =$$

We can also multiply a rational expression by a polynomial.

Example. Multiply. Simplify your answer as much as possible.

$$\frac{3x + 5}{x^3 + 27} \cdot (3x^3 - 9x^2 + 27x) =$$

$$\frac{3x + 5}{x^3 + 27} \cdot (3x^3 - 9x^2 + 27x) =$$

Dividing Rational Expressions

We divide rational expressions the same way we do fractions.

Example. Divide. Simplify your answer as much as possible.

$$\frac{3x^4y - 6x^3y - 9x^2y}{5x^2 - 20} \div \frac{3x^5y - 27x^3y}{5x^2 + 30x + 40} =$$

$$\frac{3x^4y - 6x^3y - 9x^2y}{5x^2 - 20} \div \frac{3x^5y - 27x^3y}{5x^2 + 30x + 40} =$$

We can divide polynomials in this way. Sometimes, the result is itself a polynomial.

Example. Divide. Simplify your answer as much as possible.

$$(4x^3 + 16x^2 - 9x - 36) \div (2x^2 + 5x - 12) =$$

$$(4x^3 + 16x^2 - 9x - 36) \div (2x^2 + 5x - 12) =$$

Review: Adding and Subtracting Fractions

We can add fractions with the same _____ .

Procedure: Adding or Subtracting Fractions with Like Denominator
If fractions have the same denominator, we can add or subtract them by adding or subtracting the numerators.

Example. Add or subtract. Simplify your answer as much as possible.

a. $\frac{5}{9} + \frac{7}{9} =$

b. $\frac{9}{8} - \frac{3}{8} =$

Otherwise, we have to _____ them to have the same denominator.

Procedure: Adding or Subtracting Fractions
1. Find the least common multiple of the denominators (LCD). 2. Write equivalent fractions with that denominator. 3. Add the new fractions (that have the same denominator). 4. Simplify, if necessary.

There are many methods we can use to find a common denominator with numbers. In this example, we'll use the method we'll use for rational expressions.

Example. Add $\frac{5}{6} + \frac{7}{10}$.

1. Find the LCD by looking at factors.

   $6 =$ $\qquad$ $10 =$

   LCD $=$

2. Rewrite the fractions with the LCD.

   $\frac{5}{6} =$ $\qquad$ $\frac{7}{10} =$

3. Add.

   $\frac{5}{6} + \frac{7}{10} =$

4. Simplify if necessary.

Adding and Subtracting with Like Denominators

As with fractions, we can add and subtract rational expressions with the same _____ by adding or subtracting the _____ .

Example. Add or subtract. Simplify your answer as much as possible.

a. $\dfrac{3x}{5y} + \dfrac{4x}{5y} =$

b. $\dfrac{11x^2}{4x+1} - \dfrac{7}{4x+1} =$

c. $\dfrac{x^2 - 7x + 2}{6x^2 - x - 15} + \dfrac{x^2 + 2x - 14}{6x^2 - x - 15} =$

d. $\dfrac{x^2 + 6x + 9}{x^4 - 81} - \dfrac{12x}{x^4 - 81} =$

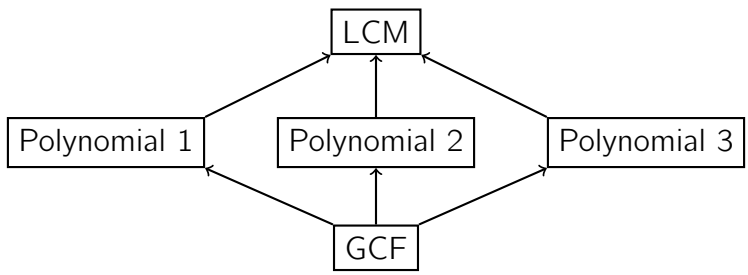
Least Common Multiple of Polynomials

To find the LCD for rational expressions, we must be able to find the least common multiple (LCM) of two (or more) polynomials.

Procedure: Finding Least Common Multiple for Polynomials

1. Factor all denominators.
2. Build the LCM:
   - Must have all factors that appear in any denominator.
   - If factors appear with different powers, take the highest power.

Note. The LCM and the GCF (greatest common factor) are found in similar ways, but the GCF must be a factor of each polynomial, and each polynomial must be a factor of the LCM.



Example. Find the least common multiple of $3x^2y$, $2x^5z$, and $9xy^3$.

LCM =

Example. Find the least common multiple of the three polynomials below. Leave the answer in factored form.

$6x^2 - 9x - 6 =$

$3x^4 - 12x^3 + 12x^2 =$

$2x^4 + x^3 - 8x^2 - 4x =$

LCM =

Adding and Subtracting Rational Expressions

To add or subtract rational expressions with different denominators, we follow the same steps as we do to add or subtract fractions of numbers:

Procedure: Adding or Subtracting Fractions
1. Find the least common multiple of the denominators (LCD). 2. Write equivalent fractions with that denominator. 3. Add the new fractions (that have the same denominator). 4. Simplify, if necessary.

Example. Add $\dfrac{x+4}{2x^2+x-6} + \dfrac{3x+1}{2x^2-13x+15}$. Simplify your answer as much as possible.

1. Find the LCD

2. Rewrite

3. Add

4. Simplify

1.10 Solving Rational Equations

Solutions to Rational Equations

Definition: Rational Equation
A **rational equation** is an equation in which both sides are rational expressions, at least one of which has a variable in the denominator.

Example. $\dfrac{3x-1}{x-2} = 4$ is a rational equation. Determine whether each value below is a solution to the equation:

a. $x = 0$ _____

b. $x = 3$ _____

c. $x = 7$ _____

Example. $\dfrac{2x-1}{x-3} = \dfrac{3x+5}{x+1}$ is a polynomial equation. Determine whether each value below is a solution to the equation:

a. $x = -2$ _____

b. $x = 0$ _____

c. $x = 2$ _____

d. $x = 7$ _____

Note that rational equations typically have _____ solution.

Clearing Denominators

We can _____ both sides of an equation by the LCD to "clear denominators."

Example. Solve $\frac{2}{3}x - \frac{4}{5} = \frac{3}{5}x + \frac{8}{15}$ by first clearing denominators.

a. Find the LCD

b. Multiply both sides by

c. Solve the new equation

d. Check

We can do the same thing when the denominators contain variables.

Example. Solve $\dfrac{x^2 - 10}{x^2} = \dfrac{3}{x}$.

a. Find the LCD

b. Multiply both sides by

c. Solve the new equation

d. Check

Extraneous Solutions

When we clear denominators in a rational equation, we multiply both sides by a _____ expression, which we've avoided so far. Why?

Example. Solve $\dfrac{3x+2}{x^2-4x-12} = \dfrac{2x+1}{x^2-2x-8}$ by clearing denominators.

a. Find the LCD

$x^2 - 4x - 12 =$ $\qquad$ $x^2 - 2x - 8 =$

LCD =

b. Multiply both sides by

c. Solve the new equation

d. Check

Wait—what happened?

If $x =$ _____, _____ $=$ _____, so we accidentally multiplied both sides by _____ . $x =$ _____ is an **extraneous solution**.

Definition: Extraneous Solution
An **extraneous solution** to an equation is one which appears through the solving process, but was not an actual solution to the equation. One reason this may occur is accidentally multiplying both sides of the equation by _____ .

Flagging Extraneous Solutions

There are two ways we can control extraneous solutions.

1. When we know one of our steps might have caused an extraneous solution, we can check our solutions in the original equation.
2. When we perform a step that might cause an extraneous solution, we can "flag" the forbidden values.

Let's see an example with the second method.

Example. Solve $\dfrac{8x-10}{x^2-3x+2} - \dfrac{x+29}{x^2+3x-4} = \dfrac{8x+2}{x^2+2x-8}$ by clearing denominators.

a. Find the LCD

$x^2 - 3x + 2 =$ $\qquad$ $x^2 + 3x - 4 =$

$x^2 + 2x - 8 =$ $\qquad$ LCD $=$

b. Multiply both sides by

$x \neq$

c. Solve the new equation

d. Check

Solve Any Equation

Solving rational equation doesn't introduce a new *technique*, but we discover a new thing that can go wrong! (New complications in red.)

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.
  - If we accidentally multiply by zero, we introduce **extraneous solutions**.
  - If we multiplied by a variable expression, we must check.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - If the function is not one-to-one, there may be many solutions.
  - If the function's range is not all real numbers, there may be no solutions.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.